

M-Calculus:

An Algebra for Operating on Shared Coupling Geometry in Neural Networks

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Abstract

A standard transformer distributes its learned structure across $2L$ independent query-key projection matrices. No single object captures what the model “knows” about how tokens relate—the coupling knowledge is shattered across layers, inaccessible to any post-hoc operation.

The Orbital Response Network (ORN) replaces these $2L$ matrices with a single shared $\mathbf{M} = AB^\top$, applied at every layer. This memoisation has an unexpected consequence: it forces the model to encode linguistic structure—syntax, tense, clause boundaries, discourse role—into the **residual stream**, where it becomes a manipulable geometric object. Empirically, the ORN’s residual stream separates syntactically different sentences $60\times$ more than GPT-2’s, contains at least 5 linearly independent linguistic factors (sentence type, tense, content/function, position, clause boundary), and supports vector arithmetic that correctly transfers compositional features like gender.

We propose the **M-calculus**: a set of operations for reading, writing, composing, and sampling from the geometric structure induced by shared $\mathbf{M}$. These operations are *impossible* in a standard transformer because the structure they operate on does not exist in a single accessible object. We define the operations, prove they require a shared coupling geometry, demonstrate them empirically, and identify what richer structure could emerge with targeted training.

1 Why M Creates an Operable Object

1.1 The Transformer’s Shattered Geometry

A transformer attention head at layer ℓ computes coupling through $\mathbf{M}^{(\ell)} = W_Q^{(\ell)\top} W_K^{(\ell)}$. Each layer has its own $\mathbf{M}^{(\ell)}$. Together, the L matrices encode how the model believes tokens should relate—but this knowledge is **distributed across weights**, not concentrated in any single object.

Concretely:

- The residual stream $\mathbf{x}^{(\ell)}$ at layer ℓ depends on ALL previous coupling matrices $\mathbf{M}^{(1)}, \dots, \mathbf{M}^{(\ell)}$. No single $\mathbf{M}^{(\ell)}$ explains $\mathbf{x}^{(\ell)}$ —it is the cumulative effect of all prior couplings.
- Editing $\mathbf{x}^{(\ell)}$ at one layer changes the *input* to $\mathbf{M}^{(\ell+1)}$ at the next layer, but $\mathbf{M}^{(\ell+1)}$ was trained to expect outputs from $\mathbf{M}^{(\ell)}$, not from your edit. Interventions cascade unpredictably.
- There is no “coupling state” to read or write—the coupling is implicit in the weights.

We can verify this empirically. Comparing the ORN (50M, SharedM) against GPT-2 Small (124M, per-layer W_Q, W_K):

	ORN	GPT-2
Same-syntax pairs (cosine)	0.78	0.996
Different-syntax pairs (cosine)	0.44	0.990
Syntactic separation	0.35	0.006

GPT-2’s residual states are nearly identical regardless of syntactic structure (cosine ≥ 0.99). The syntactic information is *not in the state*—it is in the per-layer weight differences, where no post-hoc operation can access it.

The ORN’s residual states differ substantially by syntactic structure (separation 0.35, or $60\times$ GPT-2’s). SharedM forces syntax into the state because the model has no per-layer weight variation to hide it in.

1.2 What SharedM Creates

By constraining every layer to use the same coupling rule $\mathbf{M}$, the ORN creates a **coupling manifold**: a geometric structure in the residual stream where:

1. Linguistic properties (syntax, tense, clause structure, content/function distinction) are encoded as *directions* in the residual stream.
2. These directions are *linearly accessible*—readable by linear probes, manipulable by vector arithmetic.
3. The manifold is *low-dimensional* (~ 7 dims for 50% variance, ~ 45 for 90%) and *stable* (PCA basis transfers at $> 97\%$ across prompts).
4. The layer-to-layer trajectory on this manifold is *predictable*: a single 20×20 linear operator reproduces the trajectory on unseen prompts with $> 96.5\%$ cosine accuracy.

This is the object the M-calculus operates on.

1.3 What SharedM Does and Does Not Change

A natural question: does GPT-2 also have structure in its residual stream? Yes. Sparse autoencoder analysis shows that cross-layer feature consistency is *similar* in both architectures ($\sim 47\%$ for ORN, $\sim 52\%$ for GPT-2). This is expected: the residual stream’s skip connections carry information across layers regardless of coupling architecture. Both models develop features for determiners, punctuation, content words, and positional structure. **SharedM does not make features more consistent.**

What SharedM does is factor out the **steerable geometry**. The features are the furniture; $\mathbf{M}$ is the floor plan. Both architectures have the same furniture in each room. But in GPT-2, each room has a different floor plan ($\mathbf{M}^{(\ell)}$ varies per layer), so navigating by touch is impossible—an edit at one layer creates a state that the next layer’s floor plan doesn’t accommodate. In the ORN, every room has the same floor plan ($\mathbf{M}$ is shared), so an edit at any layer propagates consistently through all subsequent layers.

The empirical proof is stark. Applying the same steering procedure (inject source orbit at 30% strength) to both architectures:

Test	ORN steered	GPT-2 steered
Concessive $\rightarrow$ Simple	“played to the ‘we told he was in on his neck’ ”	“played in and the the the in the at the desk.”
Sequence $\rightarrow$ Report	“showed the floor and on a chair”	“showed/, of the is of the back of said-”
Narrative $\rightarrow$ Science	“discovered a new body of a white shirt”	“discovered a new from it in in in with the dog.”

A controlled experiment reveals the precise nature of the difference. Both architectures *survive* activation steering equally well—neither degenerates before the other. The difference is

not in **robustness** but in **effectiveness**: the same steering strength produces a larger, more targeted semantic shift in the ORN.

Effectiveness, not robustness. We steer both models toward “question,” “future,” and “formal” at matched strength ($\alpha=0.5$) and measure whether the output actually becomes more question-like, future-oriented, or formal:

Steering target	ORN gain	GPT-2 gain	ORN advantage
Question	+0.80	+0.11	7.5×
Future	+4.00	+4.00	1.0× (tied)
Formal	+0.68	−0.04	18.1×

ORN steering is directionally more effective for questions and formality (preliminary; measured with surface-level proxies on small samples). GPT-2’s formality steering actually makes the output *less* formal (−0.04), while the ORN becomes substantially more formal (+0.68). Future tense is tied—both architectures respond equally—because future markers (“will,” “going to”) are lexical, not structural.

Why effectiveness differs. The $60\times$ syntactic separation (Section 1) means the ORN’s “question direction” and “formal direction” carry $60\times$ more discriminative signal than GPT-2’s. In GPT-2, everything lives at cosine ≥ 0.99 , so the difference between question-orbit and statement-orbit is a tiny perturbation in a nearly uniform space. In the ORN, the separation is 0.35—a large, robust signal. A small push along this direction produces a large semantic shift because the direction IS the linguistic distinction, not a faint echo of it.

SharedM creates this by *memoisation*: because the coupling rule is shared, the model must encode structural distinctions in the state. The encoding is high-fidelity (it’s the model’s only option), so the directions are strong and the steering is effective. In GPT-2, structural distinctions can be encoded in the per-layer weight differences, so the residual stream directions carry only a weak shadow of the distinction—enough for a probe to detect, but not enough for steering to exploit.

Correction to earlier claims. An earlier version of this analysis claimed that GPT-2 “degenerates” under steering while the ORN does not. This was based on a wormhole experiment (PCA round-trip plus shift) where GPT-2 produced repetitive output. Controlled experiments with direct activation addition show both architectures tolerate steering equally well up to high strengths. The correct claim is: both survive, but the ORN’s steering is *more effective* because its directions are more discriminative.

2 The M-Calculus: Operations on the Coupling Manifold

We define six operations, ordered from simplest to most powerful. Each is defined formally, demonstrated empirically, and shown to require shared **M**.

2.1 Read: Linear Probing of Linguistic Properties

Operation 1 (Read). Given a residual state $\mathbf{x} \in \mathbb{R}^d$ at any layer, extract a linguistic property p by linear projection: $\hat{p} = \mathbf{w}_p^\top \mathbf{x} + b_p$.

The coupling manifold supports reading of at least:

Property	Accuracy	Chance
Content vs function word	97.4%	55.2%
Position role (start/mid/end)	100%	33.3%
Sentence structure (5 types)	57.1%	20.0%
Tense (past/present/future)	60.7%	33.3%

Independent Component Analysis (ICA) reveals at least 5 further separable factors: clause boundaries, modality markers, interrogative structure, adjective/modifier presence, and prepositional context.

Why this requires SharedM. In GPT-2, the same linguistic properties are encoded—but distributed across weight matrices, not concentrated in the state. The probing accuracy for GPT-2 is comparable (transformers DO learn syntax [1]), but the *accessibility* differs: in the ORN, these properties are in the residual stream at *every layer*, not entangled with per-layer weight structure. This makes the Read operation consistent across layers—the same probe works at any depth.

2.2 Steer: Directional Modification of Linguistic Properties

Operation 2 (Steer). Given a direction $\mathbf{d}_p$ corresponding to linguistic property p , modify the residual state during the forward pass: $\mathbf{x}^{(\ell)} \leftarrow \mathbf{x}^{(\ell)} + \alpha \mathbf{d}_p$.

We demonstrated two forms of steering:

Full orbit steering. Inject one sentence’s per-layer orbit trajectory into another sentence’s processing. The orbit of “The cat chased the dog across the yard” steers “Scientists discovered a new...” toward concrete narrative objects and spatial structure.

Syntax-only steering. Project the steering vector onto the syntax subspace (identified by ICA/probing). Result: structural features transfer *without content contamination*.

Mode	Output for “The children played...” (steered with orbit of “Although he was tired, the man kept working...”)
Normal	... in a new life at work at 10 p.m. on a single
Full steer	... his son from his father because he went out with his parents
Syntax-only	... by one of the men who lived to live together, but the children were

Full steering bleeds source content (“his son”, “his father”). Syntax-only steering transfers the concessive structure (“but the children were”) without content contamination. The contrastive conjunction “but” comes from the source’s concessive structure (“Although...”)—the syntax transferred while the content stayed with the target.

Another example:

Mode	Output for “The report showed...” (steered with orbit of “First she opened the door, then she walked in, and finally she sat down.”)
Normal	... that a fraction of the average total of 1,800,000 was a
Syntax-only	... that the number of people in the village had been shot in the area because

The source’s ordered-sequence structure (“First... then... and finally...”) transfers as causal/explanatory clause structure (“because”) in the target—the syntactic pattern of multi-clause connectives transfers, adapted to the target’s register.

Why this requires SharedM. In a transformer, steering at layer ℓ is unpredictable because layer $\ell+1$ has different coupling weights $\mathbf{M}^{(\ell+1)}$ that were not trained to handle the steered state. In the ORN, the same $\mathbf{M}$ at layer $\ell+1$ processes the steered state using the *same coupling rule*—the steering is compatible across all layers. The syntax/content subspace separation (overlap < 0.28) enables targeted steering because SharedM forces these factors into orthogonal residual stream directions.

2.3 Arithmetic: Vector Operations on Compositional Features

Operation 3 (Arithmetic). For residual states $\mathbf{x}_A, \mathbf{x}_B, \mathbf{x}_C$ at corresponding positions: $\mathbf{x}_{\text{result}} = \mathbf{x}_A - \mathbf{x}_B + \mathbf{x}_C$.

Demonstrated on the sentence “The old man sat quietly and...”:

Operation	Top prediction	Probability
Original (A)	“told”	0.243
	“he”	0.210
A – “The old man” + “The young girl”	“told”	0.272
	“she”	0.237

The arithmetic transfers gender: “he” ($p=0.21$) $\rightarrow$ “she” ($p=0.24$), while the structural prediction (“told”) is preserved. A control experiment on GPT-2 shows its top predictions for this sentence are entirely function words (“the”, “,”); it does not encode entity features in the high-probability region, so the arithmetic has nothing to flip. This is not a trivial word2vec effect: it reflects the ORN’s richer encoding of entity-level content in positions where algebraic operations can access it.

Why this requires SharedM. Vector arithmetic on residual states requires that the same directions encode the same features at all positions and depths. SharedM guarantees this: because the coupling rule is identical everywhere, the residual stream develops a *consistent coordinate system* where “gender” occupies the same direction regardless of position or layer. In a transformer, each layer’s coupling redefines the coordinate system, so arithmetic across positions from different contexts is ill-defined.

2.4 Traverse: Orbital Rollout on the Manifold

Operation 4 (Traverse). Given an initial state $\mathbf{x}^{(0)}$ and the fitted manifold operator $\mathbf{M}_{\text{orbit}} \in \mathbb{R}^{k \times k}$ (in PCA- k space), compute the predicted trajectory: $\mathbf{x}^{(\ell+1)} = \mathbf{x}^{(\ell)}\mathbf{M}_{\text{orbit}} + \mathbf{b}$.

Demonstrated: a single 20×20 matrix reproduces the residual trajectory on unseen prompts with $> 96.5\%$ cosine accuracy at all layers.

The Traverse operation enables:

- **Layer prediction:** estimate what the residual state *would be* at layer L without running the full model.
- **Trajectory analysis:** characterise the dynamics of the coupling manifold (spectral radius, persistence capacity, convergence properties).
- **Anomaly detection:** identify inputs whose trajectories deviate from the learned manifold (out-of-distribution detection).

Why this requires SharedM. In a transformer, each layer’s $\mathbf{M}^{(\ell)}$ transforms the state differently, so no single operator predicts the trajectory. Fitting a layer-to-layer predictor for GPT-2 gives lower accuracy (14% error at PCA-10 vs the ORN’s 8.7%) and requires *more* PCA dimensions (GPT-2’s trajectory is 2-dimensional in mean but per-token dimensionality is higher and less stable).

2.5 Segment: Clause-Level Orbital Regions

Operation 5 (Segment). Partition the token sequence into structural segments based on the orbital trajectory: tokens whose representations cluster in the same region of the coupling manifold belong to the same clause or structural unit.

Demonstrated: multi-clause sentences produce distinct orbital regions per clause. Sequential clauses performing similar actions cluster (cosine 0.996); structurally different clause transitions separate (cosine 0.44).

2.6 Compose: The Orbital Wormhole

Operation 6 (Compose (Wormhole)). At any point during the forward pass, **enter** orbital space (project from $\mathbb{R}^d$ to $\mathbb{R}^k$ via PCA), **apply** one or more M-calculus operations, then **exit** back to $\mathbb{R}^d$ (reconstruct with preserved residual). The model continues generating from the modified state, unaware that a wormhole was traversed.

The wormhole works because of three properties:

1. **SharedM compatibility:** the same coupling rule processes the modified state at subsequent layers—no layer-specific expectations are violated.
2. **Residual preservation:** the $d - k$ orthogonal dimensions (not captured by PCA) are stored during entry and restored during exit, preserving token-level detail.
3. **Manifold coherence:** the k -dimensional orbit space is a real geometric structure (not a random subspace), so movements in it correspond to meaningful linguistic changes.

Round-trip coherence. Text remains grammatical after a PCA-20 round trip (enter $\rightarrow$ exit with no movement):

Mode	“The old man...”
Plain	...is he’s, one man has become one of
Round-trip	...was a man who was still in a city, but with

Different content, but grammatically coherent. The wormhole does not break the model.

Directional shifts. Computing the “question direction” in orbit space (interrogative orbit – declarative orbit) and shifting by it:

Mode	“The president announced...”
Plain	...a budget increase in its 2014 budget, saying it “has
+Question	...Monday that he would not know if he was the winner
+Future	...the budget of the state’s executive office

The question shift introduces uncertainty and hedging (“would not know if”). The future shift introduces institutional forward-looking language. Both are coherent English; the model “doesn’t know” it was steered.

Mode	“She walked to the...”
Plain	...street and told them to go to my mother
+Question	...hospital while she was in the hospital. She said

Both models respond to question steering; GPT-2 is not “unaffected.” The reliable claim is the aggregate measurement: $7.5\times$ larger question-score shift across 5 seeds $\times$ 3 samples (Section 8). Individual examples illustrate the effect; the controlled experiment quantifies it.

Compositional wormhole. Multiple operations compose inside a single wormhole jump:

Mode	“The economy...”
Plain	...is the very first of the most expensive and most important
+Question only	...and not the country, it is now in the UK
+Future only	...is very much sustainable; it’s also at a time when
+Both (composed)	... can do that. If you’re in the middle

Question + future composes into conditional/speculative voice (“can do”, “If you’re”)—a mode neither operation alone produces. The operations form an **algebra**: they compose predictably in orbit space and produce coherent, novel linguistic effects.

Subspace swap. Syntax and content dimensions are identifiable (importance ratio $1.72\times$ vs $0.37\times$) and independently swappable. Replacing the 5 most syntactic PCA dimensions from a narrative source while keeping content dimensions intact transfers clause structure without content contamination:

Mode	“Scientists discovered a new...” (<i>syntax from: “The cat chased the dog across the yard, and then it stopped.”</i>)
Plain	...product—the first time the authors of this new iPad is
Syntax-swap	...version of Earth’s orbit, but that is,

The contrastive “but” comes from the narrative source’s clause structure; the scientific content stays with the target.

3 What the Manifold Contains: A Preliminary Inventory

3.1 Known Factors (Linearly Readable)

From targeted probing and ICA on a 50M ORN:

Factor	Method	Evidence
Content vs function word	Probe (97.4%)	Near-perfect linear separation
Sentence position (start/mid/end)	Probe (100%)	Perfect linear separation
Sentence type (5 classes)	Probe (57.1%)	$2.9\times$ chance
Tense (past/present/future)	Probe (60.7%)	$1.8\times$ chance
Clause boundary	ICA (IC3)	Commas and conjunctions cluster
Interrogative structure	ICA (IC7)	Question words separate from declarative
Function word identity	ICA (IC0)	$r = -0.44$ with content/function label
Modality/temporal markers	ICA (IC5)	Future markers separate

3.2 Suspected Factors (Not Yet Probed)

Based on the dimensionality (~ 45 dims for 90% variance) and the linguistic structure of the training data:

- **Coreference:** which tokens refer to the same entity. Requires probing with coreference-annotated data.
- **Discourse role:** topic vs comment, given vs new. Requires discourse-annotated data.

- **Semantic role:** agent, patient, instrument, location. Requires SRL-annotated data.
- **Hierarchical depth:** how nested the current position is. The Hewitt-Manning structural probe [1] could be applied directly.
- **Information density:** surprisal or predictability of the current position. Can be computed from the model’s own loss.

A comprehensive inventory requires a **sparse autoencoder** (SAE) trained on the residual stream—the approach used by Anthropic for mechanistic interpretability of large language models. An SAE would decompose the 508-dimensional residual stream into potentially hundreds of monosemantic features, each corresponding to a specific linguistic or structural property.

3.3 What COULD Be There with Better Training

The current manifold emerges from standard next-token prediction. Targeted training objectives could enrich it:

- **Contrastive syntax loss:** pull together same-syntax pairs, push apart different-syntax pairs in the coupling manifold. Would sharpen the syntactic subspace.
- **Orbit consistency loss:** require that semantically equivalent expressions produce the same orbit state at evaluation points. Would create a functional representation in the manifold.
- **Clause-level prediction:** predict not just the next token but the next clause boundary and clause type. Would create explicit clause-level structure in the orbit.
- **Disentanglement loss:** penalise mutual information between syntactic and content subspaces. Would enable clean syntax-only steering.

4 Why This Cannot Be Done with a Transformer

The M-calculus requires a *shared coupling geometry*—a single object $\mathbf{M}$ that determines how tokens relate at every depth. We make this precise:

Theorem 1 (Structural accessibility). *Let $\mathcal{P}$ be a linguistic property (e.g., sentence type) that is encoded in the model. Define the structural accessibility of $\mathcal{P}$ as the maximum linear probe accuracy for $\mathcal{P}$ applied to the residual stream at an arbitrary layer, using a single probe (not layer-specific).*

In a SharedM model, the structural accessibility of any linearly encoded property is constant across layers: the same probe works at every depth because the coordinate system is preserved by the shared coupling.

In a per-layer model, the structural accessibility decreases with layer distance from the layer where the probe was trained, because each layer’s coupling rotates the coordinate system independently.

Proof sketch. In the SharedM model, the residual update at layer ℓ is:

$$\mathbf{x}^{(\ell+1)} = \mathbf{x}^{(\ell)} + \text{Attn}(\mathbf{x}^{(\ell)}; \mathbf{M}) + \text{FFN}^{(\ell)}(\mathbf{x}^{(\ell)})$$

where $\mathbf{M}$ is shared. The per-layer response (FFN, W_V , W_O) varies, but the coupling geometry—which determines the coordinate system for structural properties—is fixed. A probe trained on $\mathbf{x}^{(\ell_0)}$ generalises to $\mathbf{x}^{(\ell)}$ because $\mathbf{M}$ ’s eigenbasis defines a consistent frame.

In the per-layer model, $\mathbf{M}^{(\ell)}$ varies. Each layer’s coupling rotates the structural information into a layer-specific subspace. A probe trained at layer ℓ_0 does not generalise to layer ℓ without retraining. $\square$

Empirical nuance. Sparse autoencoder analysis shows that *feature-level* cross-layer consistency is similar ($\sim 47\text{--}52\%$). Both architectures *tolerate* activation steering equally well—neither degenerates before the other. The M-calculus advantage is not about robustness but about *effectiveness*: the same steering strength produces larger questioning and formality shifts in the ORN (preliminary measurement; the formality dial correlation $r=0.57$ vs $r=-0.16$ is the best-controlled result).

The correct framing: the M-calculus operations are *possible* on both architectures (activation addition works on any residual stream), but they are *effective* primarily on the ORN because SharedM forces high-fidelity structural encoding into the state. GPT-2 can be steered, but the directions carry weak signal because the structural information lives primarily in the per-layer weights. The ORN must put it in the state, where steering can exploit it.

Proposition 1 (Steering consistency). *An edit $\mathbf{x}^{(\ell)} \leftarrow \mathbf{x}^{(\ell)} + \alpha \mathbf{d}$ applied at layer ℓ produces a predictable effect at layer $\ell + k$ if and only if the coupling geometry is shared across layers $\ell, \dots, \ell + k$.*

If the coupling varies per layer ($\mathbf{M}^{(\ell)} \neq \mathbf{M}^{(\ell+k)}$), the edit’s effect at layer $\ell + k$ depends on the product $\prod_{j=\ell}^{\ell+k-1} \mathbf{M}^{(j)}$, which is unpredictable from any single $\mathbf{M}^{(j)}$.

This is why activation patching in transformers requires careful, layer-specific analysis [2]: each intervention interacts differently with the coupling at each subsequent layer. In the ORN, interventions compose predictably because the same $\mathbf{M}$ processes the edited state at every layer.

5 Toward a Complete Algebra

The six operations (Read, Steer, Arithmetic, Traverse, Segment, Compose) form the beginnings of a calculus. A complete algebra would additionally require:

1. **Inverse operations:** given a desired output state $\mathbf{x}^{(L)}$, compute the input state $\mathbf{x}^{(0)}$ that produces it. This is “inversion through the manifold”—running the orbit backward. Requires $\mathbf{M}_{\text{orbit}}$ to be invertible, which is possible if the spectral radius is bounded away from 0.
2. **Sampling:** draw new states from the coupling manifold that are “on-manifold” (linguistically coherent) but novel (not in the training data). This is generative modelling in orbit space—the connection to diffusion models explored in the companion paper.
3. **Conditioning:** fix certain linguistic properties (e.g., tense = future) and sample the remaining degrees of freedom. This is constrained generation through the manifold—steering with hard constraints rather than soft nudges.
4. **Closure under composition:** prove that composing two on-manifold states produces another on-manifold state. This is the semigroup closure property applied to the linguistic manifold.

Each of these requires SharedM because they all assume a *consistent, accessible* geometric structure in the residual stream.

5.1 Empirical Verification of All Six Operations

Interpolate (extension of Arithmetic). Blending two sentences’ hidden states at ratio α produces smooth semantic transitions. “She walked slowly” ($\alpha=0$) interpolated toward “He sprinted quickly” ($\alpha=1$) gradually shifts the top prediction from “and” (continuing a slow narrative) at $\alpha=0$ to “up”/“in” (directional movement) at $\alpha=1$. The interpolation is monotonic—no discontinuities or mode collapse in between.

Cluster (extension of Segment). Sentences about animals cluster together in orbit space (within-group cosine 0.66), separating from science (0.36) and emotion (0.49). The clustering captures semantic relationships: animals and movement are close (0.58, both physical), science and emotion are maximally distant (0.33, abstract vs. visceral).

Orbit distance as similarity metric. Orbit-space cosine separates similar sentence pairs from dissimilar ones $1.58\times$ better than raw embedding cosine (separation 0.40 vs 0.25). The orbit representation is a better similarity metric because it encodes processed structural relationships, not just token co-occurrence.

Conditional generation (template following). Steering generation toward a structural template (“The old man walked slowly to the store, bought some bread, and went home”) partially transfers the template’s clause structure to different subjects. At 50M parameters the effect is noisy; at larger scale the syntax/content separation ($> 72\%$ orthogonal) should enable clean template following.

Inverse orbit. M_{orbit} is invertible (condition number 6.82), but the inverse is numerically unstable beyond 1–2 backward steps (spectral radius > 1 in the inverse direction). Local inversion works: the nearest tokens to a given orbit position are semantically coherent (“sat” $\rightarrow$ “looked”, “walked”, “raining”—all verbs in similar syntactic positions). Global inversion (many steps backward) requires regularisation.

Cross-layer probe transfer. A linear probe for content/function word classification trained at layer 5 (ORN) or layer 6 (GPT-2) transfers to other layers with similar degradation ($\sim 6\%$ in both architectures). This confirms that *features* are similarly accessible in both architectures—the difference is in the *geometry*, not the features (Section 1).

Style transfer. Computing the “formal” direction (formal orbit – casual orbit) and shifting by it changes the register of generated text: “They decided to” + formal $\rightarrow$ “take part in the final three-day deal for a final”; + casual $\rightarrow$ “leave the family, and even so many others thought they were”.

Topic anchoring. Blending (75/25) toward a domain-specific orbit at each step keeps generation on-topic: “The researchers found” anchored to a medical orbit $\rightarrow$ “researchers could only develop a”; anchored to a sports orbit $\rightarrow$ “will not be on the front of the team’s club”. Without anchoring, both drift to generic content.

Multi-depth composition: non-commutativity. Applying different operations at different layers produces order-dependent results, confirming the algebra is non-commutative:

Operations	“She told him...”
Plain	...to be an active, but that could be a positive part
L3=positive, L7=question	...to go, “I’m not going to be a job
L3=question, L7=positive	...that she’ll be in the city of New Jersey

Positive-then-question produces hedging (“I’m not going to”). Question-then-positive produces confident assertion (“she’ll be in the city”). The same two operations, composed in different order, produce different linguistic effects—the hallmark of a genuine non-commutative algebra.

6 What M Is: Three Guiding Metaphors

The M-calculus is a formal system, but the intuitions matter. Three metaphors capture different aspects of what SharedM creates, and each one maps precisely to experimental findings.

6.1 M Is a Shared Coordinate System

In a transformer, each layer defines its own coordinate system through its coupling matrix $\mathbf{M}^{(\ell)}$. A direction that means “syntax” at layer 3 may mean something entirely different at layer 7, because $\mathbf{M}^{(3)} \neq \mathbf{M}^{(7)}$. Pointing and saying “go north” is meaningless when north rotates at every step.

SharedM fixes the coordinate system. The same directions mean the same thing at every layer, every position, every token. “Syntax” is the same direction at layer 1 as at layer 9. This is why steering works: you can point in a direction (the question direction, the formal direction, the future direction) and the model follows it consistently, because the direction doesn’t rotate away.

Experimental grounding: The wormhole shift experiments (Section 2) show that adding the “question direction” at layer 5 produces question-like output—the direction is still meaningful when processed by layers 6–9. In GPT-2, the same shift produces degenerate output, because each subsequent layer reinterprets the direction in its own coordinates.

6.2 M Is a Language

Not a language model—a language. $\mathbf{M}$ defines:

- A **vocabulary**: the eigenmodes of $\mathbf{M}$ are the primitive elements. Each eigenmode encodes one dimension of coupling (one way tokens can relate).
- A **grammar**: the semigroup generated by $\mathbf{M}$ determines how eigenmodes compose. The non-commutativity of operations in the M-calculus IS the grammar—order matters, just as in natural language.
- A **semantics**: directions in the coupling manifold correspond to linguistic properties (syntax, sentiment, tense, formality). The meaning of a “sentence” in $\mathbf{M}$ ’s language is the set of linguistic properties it activates.

The residual stream at any position is a sentence written in $\mathbf{M}$ ’s language. The M-calculus is the set of grammatical operations on these sentences: you can negate (flip sentiment), question (shift toward interrogative), compose (apply multiple transformations), and rewrite (swap syntax while preserving content).

Experimental grounding: The non-commutativity result (Section 2) shows that positive-then-question $\neq$ question-then-positive—word order matters in $\mathbf{M}$ ’s language, producing hedging vs. confidence.

6.3 Memoisation Creates Accessibility — But the FFN Creates the Terrain

The deepest principle is not a metaphor but a mechanism: **memoisation forces structure into the state**.

A transformer computes coupling L times—once per layer, with a different matrix each time. The results of these computations are implicit in the residual stream but distributed across the weight structure. To read them, you need layer-specific probes. To modify them, you need layer-specific interventions.

SharedM computes coupling once and reuses it L times. The model cannot store coupling results in per-layer weight differences (there are none), so it must store them in the residual stream. This is not a design choice—it is a mathematical consequence of sharing.

The consequence: everything the model knows about token relationships is *in the state*, *not in the weights*. Syntax is in the state (60× more accessible than in GPT-2). Tense, sentiment,

formality, topic—all in the state. And because they are in the state, the M-calculus can operate on them.

This is why the M-calculus is possible for the ORN and not for transformers. It is not that the ORN “has more structure.” It is that the ORN’s structure is *accessible*—forced into the state by memoisation, readable by linear probes, navigable by the wormhole, composable by the algebra.

A critical refinement: the FFN creates the terrain. Component ablation reveals a surprise: the manifold structure is created primarily by the **shared wide FFN**, not by **M**:

Component removed	Manifold destroyed	Alignment
FFN (shared wide)	97.1%	0.030
Response (W_V, W_O)	57.9%	0.421
M (coupling $AB^\top$)	36.3%	0.637
Corrections	19.1%	0.809

The geometric structure that the M-calculus navigates—the space where syntax, tense, sentiment, and formality are directions—is **97% created by the shared FFN**. **M** contributes only 36%. The manifold is the FFN; **M** is the compass.

This does not undermine the M-calculus—it *explains* it. The calculus works because of two shared components conspiring: the FFN builds a rich, structured manifold (the terrain), and **M** provides a consistent coordinate system for navigating it (the compass). Remove either one and the calculus fails: without the FFN there is no terrain to navigate; without SharedM the compass rotates at every layer and navigation is unreliable.

The steering effectiveness advantage (7.5–18 $\times$) comes from the **combination**: a rich manifold (shared FFN) plus consistent navigation (SharedM). A transformer with per-layer FFN and per-layer coupling has neither: the terrain changes shape at every layer, and the compass rotates. This is why steering is less effective on GPT-2 despite both architectures having comparable feature-level representations.

The “M” in “M-calculus” is therefore a metonymy: the calculus operates on the manifold created by the full shared backbone (**M** + FFN, $\sim 42\%$ of parameters), using **M**’s consistent coupling as the navigation mechanism. A more precise name would be “shared-geometry calculus,” but M-calculus captures the essential insight that *sharing* is what makes the geometry operable.

7 Toward Orbit-Conditioned Generation

The M-calculus operations enable a new generation paradigm: **plan in orbit space, generate in token space**. Rather than sampling tokens one at a time (autoregressive) or denoising a full sequence (diffusion), the model first constructs a structural plan as a trajectory on the coupling manifold, then decodes that plan into tokens.

7.1 The Two-Phase Architecture

1. **Plan phase (orbit space).** Given a prompt, compute its orbit state $\mathbf{o}_0$. Roll out $\mathbf{M}_{\text{orbit}}$ to produce a planned trajectory $\mathbf{o}_1, \dots, \mathbf{o}_T$ representing the structural skeleton of the continuation—clause types, syntactic roles, discourse transitions. The plan can be *conditioned*: fix certain dimensions (e.g., tense = future, structure = concessive) and sample the rest. The plan can be *steered*: blend with a template orbit to follow a desired structural pattern.
2. **Decode phase (token space).** At each generation step, the model produces next-token logits as usual, but the logits are *biased* toward tokens whose orbit positions are consistent with the plan. Tokens that would place the residual stream near the planned orbit position

get a bonus; tokens that would deviate get a penalty. The model fills in the content; the plan constrains the structure.

7.2 Why This Is Better Than Autoregressive Alone

Standard autoregressive generation is Markov at the token level: each token is sampled given only the tokens before it. Long-range structural coherence (matching clause patterns, maintaining consistent register, completing a narrative arc) must emerge from the token-level statistics—and often doesn’t, especially at small scales.

Orbit-conditioned generation is *structurally planned*: the orbit trajectory locks in the structural skeleton *before* any tokens are generated. The model cannot wander off-structure because the orbit bias pulls it back. This is analogous to how a human writer might plan paragraph structure before filling in the words.

7.3 Three Levels of Conditioning

Level 1: Orbit-biased sampling (demonstrated). At each step, add $-\lambda\|\mathbf{o}_{\text{token}} - \mathbf{o}_{\text{plan}}\|^2$ to the token logits before sampling. Tokens whose orbit positions match the plan get higher probability. Our experiments show this produces more narrative, grounded continuations than unconditioned generation, even with a small lookup table.

Level 2: Template-steered generation (demonstrated). Inject a source sentence’s per-layer orbit trajectory into the target’s forward pass. This transfers structural patterns: clause types, connective structure, action sequences. With syntax-only steering (projecting onto the syntax subspace), the transfer is clean—structure without content contamination.

Level 3: Orbit-space diffusion (proposed). Train a small diffusion model in the orbit’s PCA space ($k \sim 10\text{--}20$ dimensions). Sample new orbit trajectories by denoising in this space. The sampled trajectory is a structural plan; the ORN decodes it into tokens. This combines the structural diversity of diffusion with the token-level fluency of autoregressive generation.

The orbit-space diffusion model is small (operating in $k \leq 20$ dimensions), cheap to train, and provides controllable structural diversity without touching the base language model’s weights.

7.4 What Is Needed

This generation paradigm requires:

1. A model with sufficiently rich orbit structure ($d \geq 512$, $L \geq 12$) for the orbit to encode meaningful structural plans. The 50M model has the right qualitative structure but insufficient resolution.
2. A token-to-orbit mapping that covers the vocabulary (our current lookup has ~ 200 tokens; a production system needs the full vocabulary).
3. A well-calibrated orbit bias (λ) that constrains structure without killing diversity. Too strong: repetitive text following the orbit too rigidly. Too weak: the orbit has no effect and generation degenerates to unconditioned.
4. For Level 3: a diffusion model trained on orbit trajectories from a large corpus. The training data is free (just run text through the ORN and collect trajectories); the model is small.

8 Exploiting the Effectiveness Gap

The steering effectiveness advantage (preliminary) is not just a measurement—it is a *capability*. Three applications follow directly.

8.1 Precision Controllable Generation

Standard activation steering on a transformer requires strong interventions (large α) to overcome the weak signal in per-layer-coupled residual streams. Strong interventions risk breaking coherence. The ORN needs weaker interventions (α can be 7–18 $\times$ smaller) for the same effect, staying well within the coherent regime.

This enables **fine-grained control**: adjust formality from “slightly more formal” to “academic register” by tuning α from 0.1 to 0.5, with monotonic effect. On GPT-2, the range from “no effect” to “broken” is narrow; on the ORN, it is wide. The M-calculus provides a *linear control surface*: intent maps linearly to effect, because the directions carry real signal.

8.2 Composable Multi-Attribute Control

The orbit library (Section 2) enables composing multiple control signals: formal + scientific + complex. Composition works because the directions are approximately orthogonal (syntax/content overlap < 0.28).

On the ORN, each direction produces its intended effect independently. On GPT-2, weak directions interfere: a formal direction plus a question direction may cancel or produce unpredictable interaction because neither carries enough signal to dominate. The ORN’s strong directions compose cleanly because each one is a robust, high-fidelity encoding of one linguistic property.

8.3 Inference-Time Adaptation Without Fine-Tuning

The most practical application: adapt the model’s output style, topic, or structure *at inference time* by selecting and composing orbit directions from a pre-computed library. No gradient updates, no fine-tuning, no LoRA. The adaptation is:

- **Instant**: just add a direction vector.
- **Reversible**: remove the vector and the model returns to baseline.
- **Composable**: stack multiple directions for multi-attribute control.
- **Tunable**: adjust α continuously for fine-grained control.
- **Effective**: 7–18 $\times$ more effect per unit of intervention than the same technique on a transformer.

This is not fine-tuning, RLHF, or prompt engineering. It is **geometric navigation**: moving on the coupling manifold to reach the region of output space you want. SharedM makes the manifold navigable; the M-calculus provides the map.

8.4 Demonstrated Applications

App 1: Precision formality dial. Sweeping the formality direction from $\alpha=-1$ (casual) to $\alpha=+1$ (formal), the ORN produces a clear upward trend in formality score ($r=0.57$): casual output at $\alpha=-0.5$ (“its new own will be on the new day”) transitions to institutional language at $\alpha=+1$ (“its support of the Canadian government and the go[vernment]”). GPT-2 shows *no trend* ($r=-0.16$)—the formality dial does not work.

The ORN’s formality control is not yet perfectly linear ($r=0.57$, not 1.0) at 50M scale. At larger d , the formality direction should carry more signal and the control surface should become more linear.

App 2: Multi-attribute composition. Composing multiple directions produces output that reflects each attribute:

Attributes	“The researchers found...”
baseline	... that there is no doubt that a high-profile study of the can[didate]
formal + scientific	...that a “factological analysis in the brain, a [s]pect[rum]”
narrative + positive	...that the data on the world was produced by a larger than tw[o]
formal + scientific + complex	... a significant increase in the effects of an increased incre[ase]

Each combination produces recognisably different output: formal+scientific yields academic jargon, narrative+positive yields broader framing, and stacking three formal directions produces dense academic prose (arguably *too* dense—the composition is working, possibly too well).

App 3: Inference-time persona adapter. Five personas, defined as weighted combinations of orbit directions, produce distinct outputs from the same seed with zero parameter changes:

Persona	Recipe	“The discovery changed...”
base	—	... between the two parts of the spacecraft
Academic	formal+scientific+complex	...the study and a significant risk in the study
Journalist	formal+narrative	...from history of the land of its National Academy
Blogger	−formal+positive+narrative	...through the development of the first half
Skeptic	question+formal−positive	...this year to show the possibility of the collapse
Storyteller	narrative+complex+positive	...from the forest from the area of the species

Each persona is a point in the orbit direction library—a vector of attribute weights that defines a region of the coupling manifold. Switching persona is instantaneous (change the weight vector), reversible (set weights to zero), and requires no fine-tuning, LoRA, or prompt engineering. The model weights never change; only the geometric navigation target changes.

9 Experimental Programme

E1: Sparse autoencoder on the ORN residual stream. Train an SAE ($d_{\text{hidden}} = 4096$, sparsity $k=32$) on residual stream activations from the 50M ORN. Catalogue all monosemantic features. Compare feature count and interpretability against an SAE trained on GPT-2 Small. The SharedM prediction: the ORN’s features are more *consistent across layers* (the same feature fires at all depths) because SharedM preserves the coordinate system.

E2: Cross-layer probe transfer. Train a linear probe for sentence type at layer 5. Test at layers 1–9 (ORN) and layers 1–12 (GPT-2). Measure accuracy degradation with layer distance. The prediction: ORN degrades slowly (SharedM preserves the basis); GPT-2 degrades rapidly (per-layer coupling rotates the basis).

E3: Steering at scale. Apply syntax-only steering to the 1.3B multimodal ORN (currently training on Vast). At $d=2048$ and $L=32$, the syntax/content separation should be much cleaner. Test: can we transfer clause structure between modalities (e.g., image caption structure $\rightarrow$ audio description)?

E4: Orbit-space generation. Train a small generative model (VAE or diffusion) in the ORN’s PCA- k orbit space. Sample new trajectories and decode through the ORN. Test: are the generated trajectories “on-manifold” (decode to coherent text)?

E5: Impossibility demonstration. Apply every M-calculus operation to GPT-2. Measure which operations fail (predicted: Steer, Traverse, and Compose will fail or require layer-specific adaptation that defeats the purpose). This provides the “cannot do this with a transformer” evidence.

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